UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. What happens when an array index is outside its valid range?
   A. Undefined behaviour may occur
   B. C++ automatically resizes the array
   C. The array becomes a string
   D. The compiler always rejects the program

2. Which operator reads data from cin?
   A. =
   B. <<
   C. >>
   D. %

3. What does a compiler do?
   A. Translates the whole source program before execution
   B. Executes one source statement at a time
   C. Draws flowcharts automatically
   D. Stores only integer values

4. Which values can a bool variable hold?
   A. any decimal
   B. true or false
   C. any character
   D. any integer

5. Which declaration creates a floating-point variable named price?
   A. float price;
   B. char price;
   C. bool price;
   D. int price;

6. Which statement is most suitable for selecting among several constant cases?
   A. for
   B. do...while
   C. switch
   D. while

7. Which data type stores a whole number?
   A. string
   B. float
   C. char
   D. int

8. Which statement immediately exits a loop?
   A. continue
   B. return 0
   C. cin
   D. break

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9. How many times does a while loop execute when its condition is false initially?
   A. Until the user stops it
   B. Two
   C. One
   D. Zero

10. Which term describes a finite sequence of precise steps that solves a problem?
   A. Variable
   B. Compiler
   C. Algorithm
   D. Array

PART B (60 MARKS)

QUESTION 1

This question tests algorithms and programming errors.

a) Define algorithm and pseudocode.

(2 marks)

b) Explain THREE types of programming errors and give one example of each.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade: 80-100 A, 60-79 B, 50-59 C, 40-49 D, below 40 F.

a) Draw a flowchart for the grade classification.

(8 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(5 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(6 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(5 marks)

QUESTION 4

Study the following C++ program segment:

```
int total = 0;
for (int i = 1; i <= 5; i++) {
   total += i;
}
cout << total;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

(9 marks)

QUESTION 5

A sports club stores sport names and medal counts in parallel arrays.

a) Declare two parallel arrays for five sports and initialise them with suitable values.

(6 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(10 marks)

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PART C (30 MARKS)

QUESTION 1

A cafe sells three items: coffee RM8.50, tea RM6.00 and juice RM9.50. A 10% discount is given when the subtotal is at least RM50.

Write a complete C++ program that reads the item code and quantity, calculates the subtotal and discount, and displays a formatted receipt with the item name, quantity, subtotal, discount and amount to pay.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER